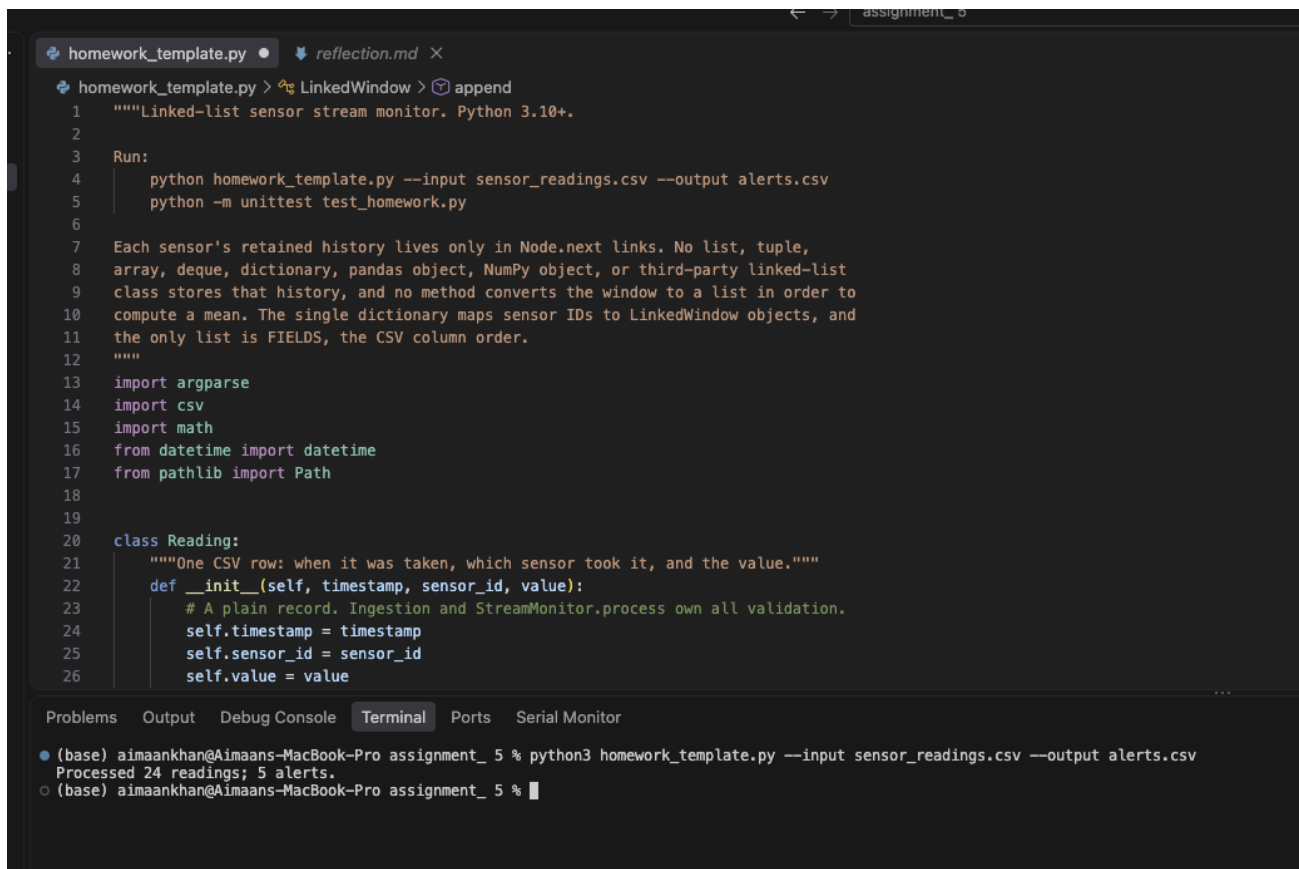


Test Evidence: Sensor Stream Monitor

Command:

```
python3 homework_template.py --input sensor_readings.csv --output alerts.csv
```

Terminal Output:



The screenshot shows a code editor with a file named `homework_template.py` and a terminal window below it. The code editor displays the following Python code:

```
1 """Linked-list sensor stream monitor. Python 3.10+.  
2  
3 Run:  
4     python homework_template.py --input sensor_readings.csv --output alerts.csv  
5     python -m unittest test_homework.py  
6  
7 Each sensor's retained history lives only in Node.next links. No list, tuple,  
8 array, deque, dictionary, pandas object, NumPy object, or third-party linked-list  
9 class stores that history, and no method converts the window to a list in order to  
10 compute a mean. The single dictionary maps sensor IDs to LinkedWindow objects, and  
11 the only list is FIELDS, the CSV column order.  
12 """  
13 import argparse  
14 import csv  
15 import math  
16 from datetime import datetime  
17 from pathlib import Path  
18  
19  
20 class Reading:  
21     """One CSV row: when it was taken, which sensor took it, and the value."""  
22     def __init__(self, timestamp, sensor_id, value):  
23         # A plain record. Ingestion and StreamMonitor.process own all validation.  
24         self.timestamp = timestamp  
25         self.sensor_id = sensor_id  
26         self.value = value
```

The terminal window shows the output of the command:

```
(base) aimaankhan@Aimaans-MacBook-Pro assignment_5 % python3 homework_template.py --input sensor_readings.csv --output alerts.csv  
Processed 24 readings; 5 alerts.  
(base) aimaankhan@Aimaans-MacBook-Pro assignment_5 %
```

File output:

```
alerts.csv > data
1 timestamp,sensor_id,value,baseline_mean,abs_deviation,status>window_size_after,evicted_value
2 2026-09-01T09:00:00,TEMP_A,20.00,,WARMUP,1,
3 2026-09-01T09:00:00,TEMP_B,30.00,,WARMUP,1,
4 2026-09-01T09:01:00,TEMP_A,21.00,,WARMUP,2,
5 2026-09-01T09:01:00,TEMP_B,31.00,,WARMUP,2,
6 2026-09-01T09:02:00,TEMP_A,19.00,,WARMUP,3,
7 2026-09-01T09:02:00,TEMP_B,29.00,,WARMUP,3,
8 2026-09-01T09:03:00,TEMP_A,20.00,20.00,0.00,OK,4,
9 2026-09-01T09:03:00,TEMP_B,36.00,30.00,6.00,ALERT,4,
10 2026-09-01T09:04:00,TEMP_A,28.00,20.00,8.00,ALERT,5,
11 2026-09-01T09:04:00,TEMP_B,30.00,31.50,1.50,OK,5,
12 2026-09-01T09:05:00,TEMP_A,21.00,21.60,0.60,OK,5,20.00
13 2026-09-01T09:05:00,TEMP_B,31.00,31.20,0.20,OK,5,30.00
14 2026-09-01T09:06:00,TEMP_A,20.00,21.80,1.80,OK,5,21.00
15 2026-09-01T09:06:00,TEMP_B,22.00,31.40,9.40,ALERT,5,31.00
16 2026-09-01T09:07:00,TEMP_A,19.00,21.60,2.60,OK,5,19.00
17 2026-09-01T09:07:00,TEMP_B,30.00,29.60,0.40,OK,5,29.00
18 2026-09-01T09:08:00,TEMP_A,12.00,21.60,9.60,ALERT,5,20.00
19 2026-09-01T09:08:00,TEMP_B,31.00,29.80,1.20,OK,5,36.00
20 2026-09-01T09:09:00,TEMP_A,20.00,20.00,0.00,OK,5,28.00
21 2026-09-01T09:09:00,TEMP_B,30.00,28.80,1.20,OK,5,30.00
22 2026-09-01T09:10:00,TEMP_A,21.00,18.40,2.60,OK,5,21.00
23 2026-09-01T09:10:00,TEMP_B,38.00,28.80,9.20,ALERT,5,31.00
24 2026-09-01T09:11:00,TEMP_A,22.00,18.40,3.60,OK,5,20.00
25 2026-09-01T09:11:00,TEMP_B,30.00,30.20,0.20,OK,5,22.00
26
```